

My AoPs Solution Intermediate Algebra

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1 Substitution Exercises

1. Find all ordered pairs (x, y) such that $-x + 2y = 8$ and $x - y = -5$

$$y = x + 5$$

Substituting into the first equation we get $-x + 2(x + 5) = 8$

$$x + 10 = 8$$

$$x = -2$$

Substituting into the first equation we get $2 + 2y = 8$

$$\therefore y = 3$$

$$(x, y) = (-2, 3)$$

2. Find b if a is constant such that $3ax - 8b = 4x + 6a$ for all values of x . Since the equation is true for all values of x it means that x can be set to 0. Thus

$$-8b = 6a$$

It also means that $x = 1$. $\therefore 3a - 8b = 4 + 6a$. Since $-8b = 6a$

$$3a + 6a = 4 + 6a$$

$$3a = 4 \therefore a = \frac{4}{3}$$

Now that we have a we are able to find b .

$$-8b = 6a$$

$$-8b = 8 \therefore b = -1$$

3. Find all ordered pairs (x, y) such that $x + y = 6$ and $x^2 + xy + 2y^2 = 32$. From the first equation we get that $x = 6 - y$. By substitution into the second equation we get

$$(6 - y)^2 + y(6 - y) + 2y^2 = 32$$

$$36 - 12y + y^2 + 6y - y^2 + 2y^2 = 32$$

$$2y^2 - 6y + 4 = 0$$

$$y^2 - 3y + 2 = 0$$

$$(y - 1)(y - 2) = 0$$

$\therefore y = 1, 2$ Then substitute y into the first equation $x + y = 6$ To get that $x = 5, 4$. Therefore the ordered pairs are $(4, 2), (5, 1)$.

4. Solve the system of equations:

$$x + 2y - z = 5$$

$$3x + 2y + z = 11$$

$$(x + 2y)^2 - z^2 = 15$$

Here I can identify the $x + 2y$ in equation 1 and 3. $x + 2y = 5 + z$

$$(5 + z)^2 - z^2 = 15 \quad (1)$$

$$25 + 10z + z^2 - z^2 = 15 \quad (2)$$

$$10z + 25 = 15 \quad (3)$$

$$10z = -10, \therefore z = -1 \quad (4)$$

Now that we have z we can substitute it into the other equations.

$$x + 2y + 1 = 5 \quad x + 2y = 4 \quad (5)$$

$$3x + 2y - 1 = 11 \quad 3x + 2y = 12 \quad (6)$$

Now the last equation can be rewritten as $2x + (x + 2y) = 12$ $2x + 4 = 12 \therefore x = 4$ Now we substitute x back into the first equation to get $4 + 2y = 4 \therefore y = 0$.

2 Elimination Exercises

- Find all ordered pairs (a, b) such that $3a - 2b = -8$ and $5a + 4b = 5$

Multiply the first equation by 2 so that $-2b$ becomes $-4b$, resulting in an easy addition of the two equations.

$$6a - 4b = -16$$

Now add the two equations and we get

$$11a = -11$$

Thus $a = -1$. Substitute back into the first equation to get

$$-3 - 2b = -8 \therefore -2b = -5, b = \frac{5}{2}$$

Thus the ordered pair is $(a, b) = \left(-1, \frac{5}{2}\right)$.